

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2022/2023

MAY EXAMINATION

UCCD2003 OBJECT-ORIENTED SYSTEMS ANALYSIS AND DESIGN

MONDAY, 22 MAY 2023

TIME : 2.00 PM – 4.00 PM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING

Instruction to Candidates :

This question paper consists of **THREE (3)** questions in Section A and **TWO (2)** questions in Section B.

Answer **ALL** questions in Section A and **ONLY ONE (1)** question in Section B. Each question carries 25 marks.

Should a candidate answer more than **ONE (1)** question in Section B, marks will only be awarded for the **FIRST** question in the order the candidate submits the answers.

Candidates are not allowed to use a calculator.

Answer questions only in the answer booklet provided.

UCCD2003 OBJECT-ORIENTED SYSTEMS ANALYSIS AND DESIGN**Section A (Compulsory Questions)**

- Q1. (a) List **SIX (6)** types of factors used for selecting a methodology to build a system. (6 marks)
- (b) Discuss **FOUR (4)** major types of non-functional requirements used in system modeling during the design phases. (8 marks)
- (c) Assume you are a project manager. Suggest **TWO (2)** ways to increase the motivation of the project team to improve the performance of system development. (4 marks)
- (d) What is the major difference between phased development and throwaway prototyping? (3 marks)
- (e) Explain **TWO (2)** types of activities used in business requirement improvement to do requirement analysis. (4 marks)
- [Total : 25 marks]

- Q2. (a) A customer can select products and add them to the shopping cart. If the customer is not a member, he/she is required to enter his/her nickname, mailing address and contact number. Otherwise, the customer is required to sign in into his/her account with a username and password. The shopping cart must have at least one or more products before the customer can perform the checkout. Each product has a product description, id, price and etc. The purchase order will be initiated for the customer after he/she has checkout the shopping cart. The customer can make the payment using three different types of payment methods: cash, credit card or e-wallet. All the payment information will be recorded into the system database. Given the mentioned scenario, use the class diagram notation to draw the design of data structure with items listed below.
- (i) Association relationships between two classes. (8 marks)
- (ii) Generalization relationships between two classes. (3 marks)
- (iii) Aggregation relationships between two classes. (2 marks)
- [Hints: Combine the answer (i), (ii) and (iii) into a class diagram]
- (b) Discuss the differences among attribute, aggregation and composition used for the class diagram. (6 marks)

UCCD2003 OBJECT-ORIENTED SYSTEMS ANALYSIS AND DESIGN**Q2. (Continued)**

- (c) Explain **THREE (3)** ways to optimize the design of class and method to ensure it balances understandability with efficiency. (6 marks)

[Total : 25 marks]

- Q3. (a) An e-wallet system has three types of user accounts: Lite account, Pro account and Premium account. The monthly transaction limits for Lite account, Pro account and Premium account are RM2,000.00, RM5,000.00 and RM25,000.00, respectively. The e-wallet allows users to transfer money through DuitNow, reload money to the e-wallet and withdraw money to their own bank accounts. In addition, a user can make payment and/or receive payment using either generating or scanning the barcode or QR code. The generated barcode and QR code will expire and become invalid after 60 seconds. This will cause a user to be unable to make payments and/or receive payments successfully. The user also will not be able to make any payment, if the total amount of monthly transactions exceeds the limit. Given the mentioned scenario, use the use case diagram notation to draw the system functionalities with items listed below.

- (i) Actor(s) who interact with the system. (1 mark)
 - (ii) Use case that associated with actor(s) using association relationship(s). (5 marks)
 - (iii) Use case that associated with others use case(s) using either include or extend relationship(s). (4 marks)
 - (iv) Generalization relationship(s). (3 marks)
 - (v) System boundary. (1 mark)
- [Hints: Combine the answer (i), (ii), (iii), (iv) and (v) into a use case diagram]

UCCD2003 OBJECT-ORIENTED SYSTEMS ANALYSIS AND DESIGN**Q3. (Continued)**

(b) Given the flow of events,

- I. The customer checkout the gift redemption request.
- II. The system checks whether the company has sufficient stock to redeem and the customer has sufficient points to make the redemption at the same time.
The system will reject the redemption request, if the company has insufficient stock or the customer has insufficient points. Otherwise, proceed to next.
- III. The system informs the redemption department to prepare the gift.
- IV. The redemption department ships the gift to the customer.
- V. The customer confirms receipt of the gift through the system.
- VI. The system deducts the points from the customer account.

Based on the flow of events given, use the activity diagram with swimlane notation to draw the gift redemption process with items listed below.

- (i) Swimlane. (3 marks)
- (ii) Step-by-step activities of the gift redemption process. (3 marks)
- (iii) Decision nodes used for the gift redemption process. (3 marks)
- (iv) Parallel process used for the gift redemption process. (1 mark)
- (v) The usage of initial node and final-activity node. (1 mark)

[Hints: Combine the answer (i), (ii), (iii), (iv and (v) into an activity diagram with swimlane]

[Total : 25 marks]

UCCD2003 OBJECT-ORIENTED SYSTEMS ANALYSIS AND DESIGN**Section B (Choose any ONE Question)**

Q4. (a) Given the flow of events,

- I. An instructor activates the class attendance session.
- II. The system checks the activation time. The class attendance session will only be successfully activated if the activation time does not exceed the set class attendance session timeout. Otherwise, the system prompts a “activation is failed” message to the instructor.
- III. The system generates a QR code.
- IV. A student scans the QR code.
- V. The system verifies the QR code. If the QR code is invalid, the system prompts a “QR invalid” message to the student. Otherwise, proceed to next.
- VI. The system checks the availability of the student in the class attendance list. If the student exists in the attendance list, the system updates the student attendance record. Otherwise, the system places the record into attendance pending queue.
- VII. The system deactivates the class attendance session automatically after end of the set class attendance session timeout.

Based on the flow of events given, use the sequence diagram notation to draw the attendance taking process with items listed below.

- (i) Actor(s) and Object(s). (3 marks)
 - (ii) Message passing. (3 marks)
 - (iii) Guard condition message passing. (6 marks)
 - (iv) Lifeline and execution occurrence components. (2 marks)
- [Hints: Combine the answer (i), (ii), (iii) and (iv) into a sequence diagram]
- (b) Discuss **TWO (2)** ways to balance the analysis model to ensure the consistency among the unified modeling language (UML) diagrams produced at the end of the analysis phase. (4 marks)
 - (c) Explain in which situation, you will use custom development approach as the design strategies to build the system, compared to the outsourcing approach. (2 marks)
 - (d) Identify **FIVE (5)** types of object persistence format used for data storage design. (5 marks)

[Total : 25 marks]

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- Q5. (a) An e-wallet system allows a user to make payment, receive payment and check history transactions. Given the mentioned scenario, use the window navigation diagram notation to draw the structure user interface design with items listed below.
- (i) Box(es) with appropriate stereotype that represent(s) the user interface(s). (5 marks)
 - (ii) Transition paths between the interfaces. (2 marks)
[Hints: Combine the answer (i) and (ii) into a window navigation diagram]
- (b) Identify **SIX (6)** types of principles used for a good user interface design. (6 marks)
- (c) Explain the differences between server-based architecture and client-server architecture using software architectural components and hardware architectural components in terms of physical architectural design. (6 marks)
- (d) Explain the **THREE (3)** types of user documentations to be prepared at the end of the implementation phase. (6 marks)
[Total : 25 marks]
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